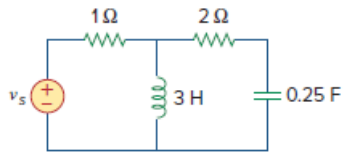


Problem

Assuming that $v_s = 8 \cos(2t - 40^\circ)$ V in the circuit of Fig. 11.37, find the average power delivered to each of the passive elements.

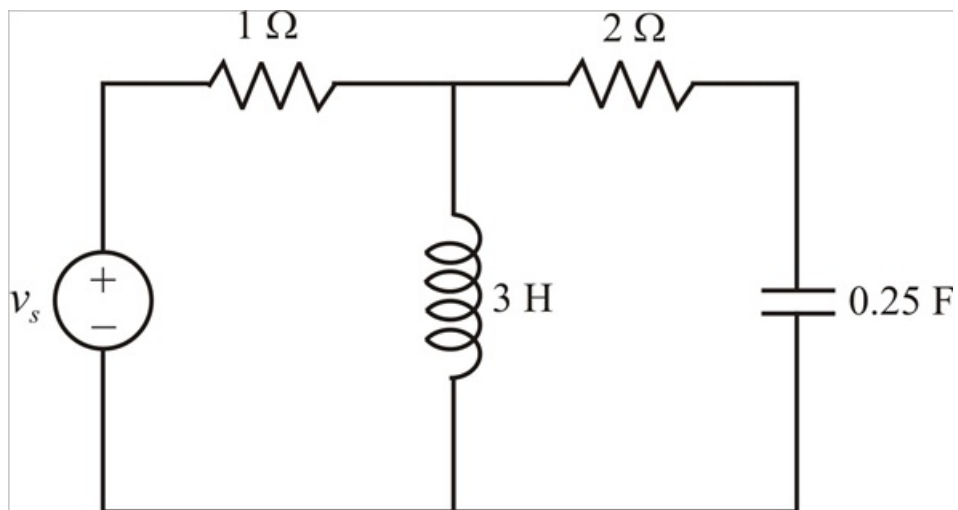
Figure 11.37:



Step-by-step solution

Step 1 of 11

Given circuit:



Step 2 of 11

Given data is

Source voltage $v_s = 8 \cos(2t - 40^\circ)$ V

We need to find the average power delivered to each of passive elements.

Step 3 of 11

To convert the given circuit into frequency-domain circuit, we need to do the following calculations:

Source voltage $\mathbf{V}_s = 8\angle -40^\circ \text{ V}$

Angular frequency $\omega = 2 \text{ rad/s}$

Inductive reactance is given by

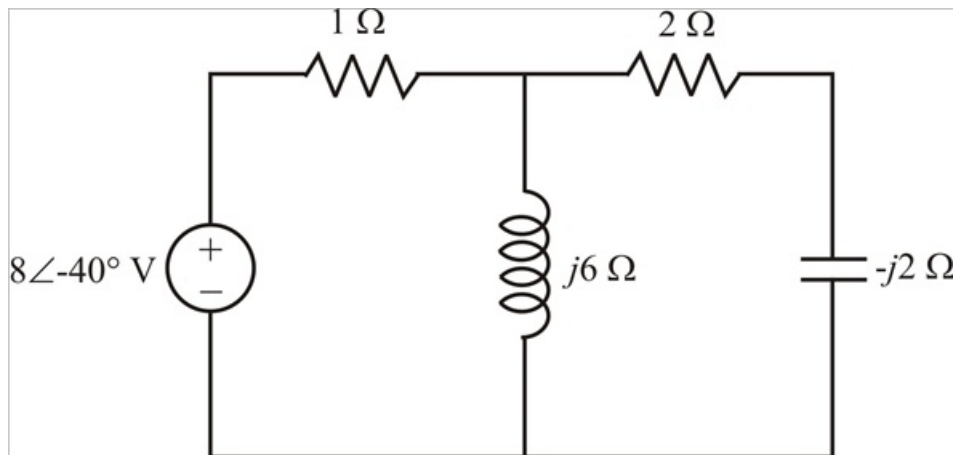
$$\begin{aligned}X_L &= j\omega L \\&= j(2)(3) \\X_L &= j6\Omega\end{aligned}$$

Capacitive reactance is given by

$$\begin{aligned}X_C &= \frac{1}{j\omega C} \\&= \frac{1}{j(2)(0.25)} \\X_C &= -j2\end{aligned}$$

Step 4 of 11

Now the circuit in frequency-domain is



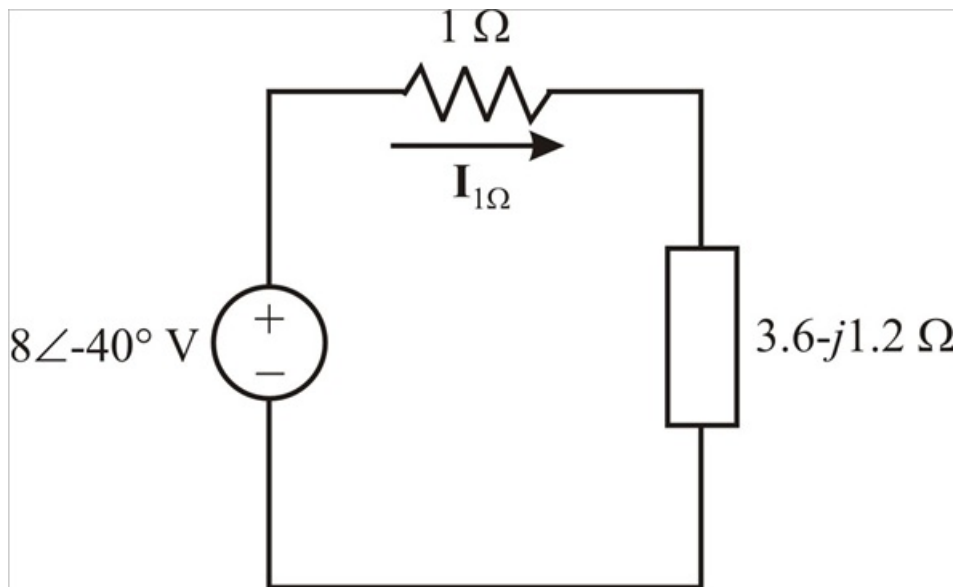
Step 5 of 11

In the above circuit, series combination of 2Ω and $-j2\Omega$ is in parallel with $j6\Omega$. Hence its equivalent impedance is given by

$$\begin{aligned}&\Rightarrow (2 - j2) \parallel (j6) \\&\Rightarrow \frac{(2 - j2)(j6)}{2 - j2 + j6} \\&\Rightarrow 3.6 - j1.2\end{aligned}$$

Step 6 of 11

Now the above circuit is modified as shown below



Step 7 of 11

By applying Kirchhoff's voltage law to the above circuit, we get

$$-8\angle -40^\circ + (1 + 3.6 - j1.2) \mathbf{I}_{1\Omega} = 0$$

$$(1 + 3.6 - j1.2) \mathbf{I}_{1\Omega} = 8\angle -40^\circ$$

$$\begin{aligned} \mathbf{I}_{1\Omega} &= \frac{8\angle -40^\circ}{4.6 - j1.2} \\ &= 1.52 - j0.72 \end{aligned}$$

$$\mathbf{I}_{1\Omega} = 1.6828\angle -25.38^\circ \text{ A}$$

Therefore, the current through 1Ω resistance is $\mathbf{I}_{1\Omega} = 1.6828\angle -25.38^\circ \text{ A}$

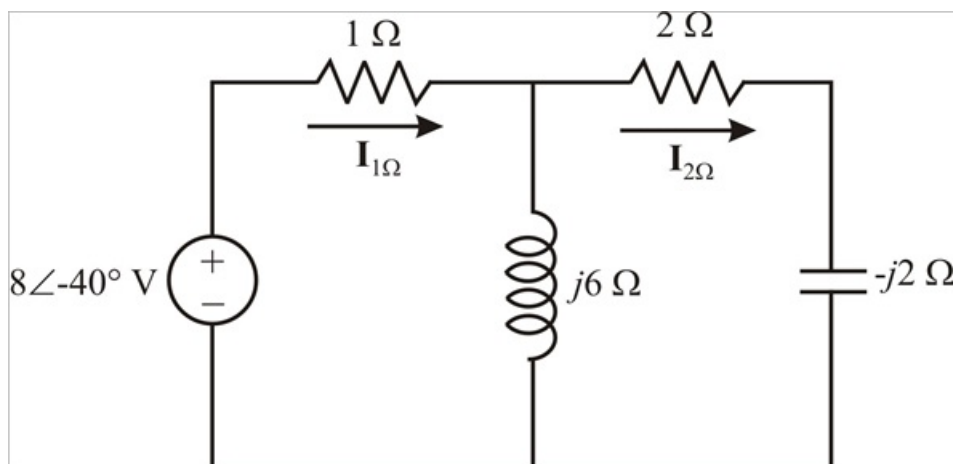
Average power delivered to 1Ω resistor is given by

$$\begin{aligned} P_{1\Omega} &= \frac{1}{2} |\mathbf{I}_{1\Omega}|^2 R \\ &= \frac{1}{2} (1.6828)^2 (1) \\ P_{1\Omega} &= 1.4159 \text{ W} \end{aligned}$$

Therefore, the average power delivered to 1Ω resistance is $P_{1\Omega} = 1.4159 \text{ W}$.

Step 8 of 11

Consider the following circuit to find the current through 2Ω resistor.



Step 9 of 11

By applying current division to above circuit, we get

$$\begin{aligned} \mathbf{I}_{2\Omega} &= \left(\frac{j6}{2 - j2 + j6} \right) \mathbf{I}_{1\Omega} \\ &= (1.2 + j0.6)(1.6828 \angle -25.38^\circ) \\ &= 2.257 + j0.0467 \\ \mathbf{I}_{2\Omega} &= 2.257 \angle 1.185^\circ \text{ A} \end{aligned}$$

Step 10 of 11

Average power delivered to 2Ω resistor is given by

$$\begin{aligned} P_{2\Omega} &= \frac{1}{2} |\mathbf{I}_{2\Omega}|^2 R \\ &= \frac{1}{2} (2.257)^2 (2) \\ P_{2\Omega} &= 5.09 \text{ W} \end{aligned}$$

Therefore, the average power delivered to 2Ω resistance is $P_{2\Omega} = 5.09 \text{ W}$.

Step 11 of 11

We know that a pure reactive element absorbs no average power. Hence inductor and capacitor in the given circuit absorbs zero average power. Therefore

$$\begin{aligned} P_{3H} &= 0 \text{ W} \\ P_{0.25F} &= 0 \text{ W} \end{aligned}$$